

Developing Research Based Web Applications Using Red Hat OpenShift



OpenShift is a Kubernetes based container application, designed for container based software deployment and management. It is an application development and hosting platform, which automates management and enables the developer to focus on the app itself.

The increase in the volume, velocity and the variety of data from multiple channels demands high performance computing resources that can process heterogeneous Big Data. It is not always possible to purchase costly computing resources like high performance multi-core processors with supercomputing powers, huge memory devices and related technologies to process, visualise and make predictions on the datasets related to live streaming and real-time supercomputing applications. To cope with and work with such technologies, cloud services are used, whereby computing resources can be hired on demand and billed as per usage.

There are a number of cloud services providers in the global market with different delivery models including Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS) and Software-as-a-Service (SaaS). Nowadays, there are some new keywords in the cloud delivery space, like Network-as-a-Service (NaaS), Database-as-a-Service (DBaaS), Testing-as-a-Service (TaaS) and many others. Each of these cloud delivery approaches has different resources, which are used for different applications.

Features of Red Hat OpenShift

Red Hat OpenShift is one of the leading cloud services providers in the PaaS (Platform as a Service) paradigm. It provides multiple platforms to cloud users with the flexibility to develop, deploy and execute applications on the cloud. OpenShift has high performance data centres with enormous processing power to work with different programming languages, which include Java, PHP, Ruby, Python, Node.js, Perl, Jenkins Server, Ghost, Go and many others.

A beginner can use the Free Tier of Red Hat OpenShift for the development, deployment and execution of new cloud apps on the online platform provided by it. Any of the programming languages mentioned can be used for the development of apps with real-time implementation.

Developing PHP research based Web applications

OpenShift provides multiple programming language options to cloud users for the development of apps. With each programming language, OpenShift delivers multiple versions so that the compatibility issues can be avoided at later stages.